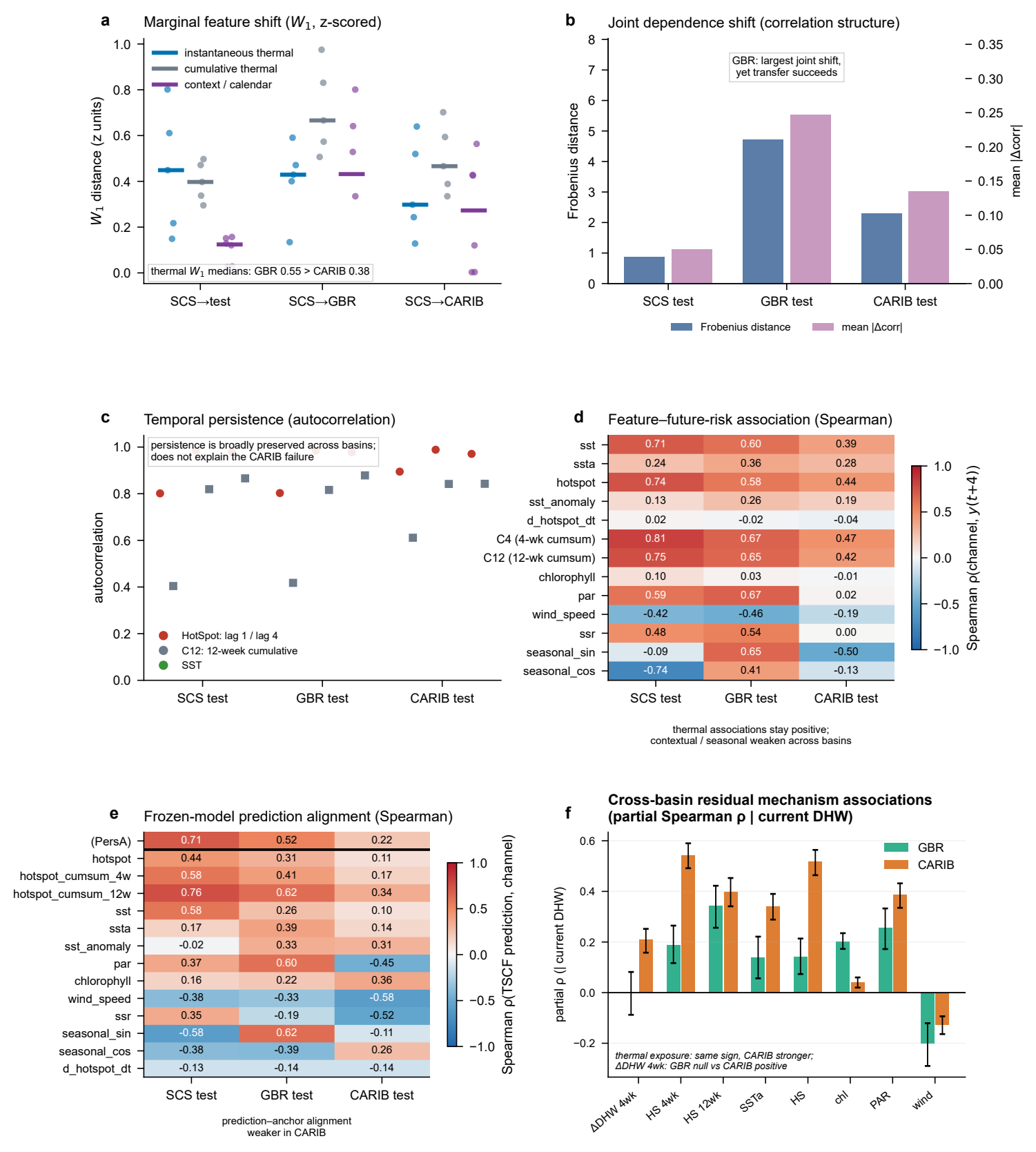

a
Marginal feature shift (W_1, z-scored)
instantaneous thermal
cumulative thermal
context / calendar
W_1 distance (z units)
thermal W_1 medians: GBR 0.55 > CARIB 0.38
SCS→test
SCS→GBR
SCS→CARIB
b
Joint dependence shift (correlation structure)
GBR: largest joint shift,
yet transfer succeeds
Frobenius distance
mean |Δcorr|
SCS test
GBR test
CARIB test
Frobenius distance
mean |Δcorr|
c
Temporal persistence (autocorrelation)
persistence is broadly preserved across basins;
does not explain the CARIB failure
autocorrelation
HotSpot: lag 1 / lag 4
C12: 12-week cumulative
SST
SCS test
GBR test
CARIB test
d
Feature–future-risk association (Spearman)
sst 0.71 0.60 0.39
ssta 0.24 0.36 0.28
hotspot 0.74 0.58 0.44
sst_anomaly 0.13 0.26 0.19
d_hotspot_dt 0.02 -0.02 -0.04
C4 (4-wk cumsum) 0.81 0.67 0.47
C12 (12-wk cumsum) 0.75 0.65 0.42
chlorophyll 0.10 0.03 -0.01
par 0.59 0.67 0.02
wind_speed -0.42 -0.46 -0.19
ssr 0.48 0.54 0.00
seasonal_sin -0.09 0.65 -0.50
seasonal_cos -0.74 0.41 -0.13
SCS test
GBR test
CARIB test
Spearman ρ(channel, y(t+4))
thermal associations stay positive;
contextual / seasonal weaken across basins
e
Frozen-model prediction alignment (Spearman)
(PersA) 0.71 0.52 0.22
hotspot 0.44 0.31 0.11
hotspot_cumsum_4w 0.58 0.41 0.17
hotspot_cumsum_12w 0.76 0.62 0.34
sst 0.58 0.26 0.10
ssta 0.17 0.39 0.14
sst_anomaly -0.02 0.33 0.31
par 0.37 0.60 -0.45
chlorophyll 0.16 0.22 0.36
wind_speed -0.38 -0.33 -0.58
ssr 0.35 -0.19 -0.52
seasonal_sin -0.58 0.62 -0.11
seasonal_cos -0.38 -0.39 0.26
d_hotspot_dt -0.13 -0.14 -0.14
SCS test
GBR test
CARIB test
Spearman ρ(TSCF prediction, channel)
prediction–anchor alignment
weaker in CARIB
f
Cross-basin residual mechanism associations
(partial Spearman ρ | current DHW)
GBR
CARIB
partial ρ (| current DHW)
thermal exposure: same sign, CARIB stronger;
ΔDHW 4wk: GBR null vs CARIB positive
ΔDHW 4wk
HS 4wk
HS 12wk
SSTa
HS
chl
PAR
wind